Sturktur tabel

```sql
CREATE TABLE users (
  id INT PRIMARY KEY,
  first_name VARCHAR(255),
  last_name VARCHAR(255),
  age INT,
  gender VARCHAR(50),
  country VARCHAR(100),
  register_at TIMESTAMP
);
CREATE TABLE products (
  id INT PRIMARY KEY,
  cost NUMERIC(15,8),
  category VARCHAR(255),
  name VARCHAR(255)
);
CREATE TABLE order_details (
  id INT PRIMARY KEY,
  order_id INT,
  user_id INT,
  product_id INT,
  quantity INT,
  status VARCHAR(100),
  created_at TIMESTAMP,
  shipped_at TIMESTAMP,
  delivered_at TIMESTAMP,
  returned_at TIMESTAMP,
  sale_price NUMERIC(15,8)
)
```

1. produk mengandung “Sport”

```sql
select * from products
where name like '%sport%'
order by id asc
limit 10
```

Hasil

	id	cost	category	name
1	2,813	10.09595993	Active	Woman Within Plus Size Petite pants in sports knit
2	2,919	10.27088996	Active	Woman Within Plus Size Tall pants in sports knit
3	3,143	18.78910022	Dresses	2B Passport Stripe High Low Dress
4	9,809	13.96940988	Sleep & Lounge	Violet and Black tartan plaid check tie cord flannel cotton pants for lounging sleep sports. Unisex relaxed fit
5	10,025	15.5937599	Sleep & Lounge	Electric Blue and Black tartan plaid check tie cord flannel cotton pants for lounging sleep sports. Unisex relaxed fit
6	10,604	10.18799996	Intimates	Fruit of the Loom 3-Pack Racerback sport bras
7	13,738	8.8955499	Accessories	Leather Zipper Travel Credit Card Passport Wallet Available in Different Colors
8	13,904	5.31674999	Accessories	Travel Passport Boarding Pass Ticket Wallet Available in Different Colors
9	14,264	23.62394915	Accessories	Lodis Audrey Passport Wallet with Ticket Flap
10	15,490	10.3148399	Plus	Leather Zipper Travel Credit Card Passport Wallet Available in Different Colors

2. 10 produk paling mahal & 10 paling murah

```
(select name, cost from products
        order by cost desc
        limit 10)
union all
(select name, cost from products
        order by cost asc
        limit 10)
```

Hasil :

	name	cost
1	Nike Jordan Retro 11 Bred Bootie Socks	557.1510022
2	Jordan Low Quarter Sock Style # 427411	537.2850014
3	The North Face Apex Bionic Soft Shell Jacket - Men's	524.6430002
4	NIKE WOMEN'S PRO COMPRESSION SPORTS BRA *Outstanding Support and Comfort*	512.0009998
5	Alpha Industries Rip Stop Short	482.5169991
6	Woolrich Arctic Parka DF	478.1700028
7	AIR JORDAN DOMINATE SHORTS MENS 465071-100	454.209
8	The North Face Denali Down Womens Jacket 2013	437.052001
9	The North Face Denali Down Mens Jacket 2013	436.1490025
10	Catherine Malandrino Women's Skinny Stretch Leather Pant	434.0749978
11	Indestructable Aluminum Aluma Wallet - RED	0.0083
12	Set of 2 - Replacement Insert For Checkbook Wallets Card Or Picture Insert	0.17738
13	Wayfarer Style Sunglasses Dark Lens Black Frame	0.606
14	GENUINE LEATHER SNAP ON STUDDED WHITE PIANO BELT FITS ANY BUCKLE	0.618
15	Solid Color Leather Adjustable Skinny Belt with	0.64476999
16	Classic Tear Drop Mirror Lens Aviator Sunglasses	0.64500001
17	Blank Long Cuff Beanie Cap (Choose Many Colors Available)	0.74984002
18	Pink Ribbon Breast Cancer Awareness Knee High Socks Great for Sports Teams Fundrais	0.76830002
19	Pink Ribbon Breast Cancer Awareness Knee High Socks Great for Sports Teams Fundrais	0.77220002
20	Motherhood Maternity: Bra Back Extender	0.78408001

3. Total jumlah barang per kategori di negara yang berada di benua Asia

Indentifikasi Negara Asia

```
select distinct country from users;
```

```
select t2.category, count(t2.cost) as quantity
from order_details t1
left join products t2
  on t1.product_id = t2.id
left join users t3
  on t1.user_id = t3.id
where (t1.status = 'Complete') and (t3.country in ( 'China', 'Japan', 'South Korea'))
group by t2.category
```

Hasil :

	category	quantity
1	Maternity	190
2	Accessories	406
3	Leggings	139
4	Jumpsuits & Rompers	40
5	Clothing Sets	9
6	Intimates	552
7	Skirts	88
8	Swim	461
9	Socks	270
10	Sweaters	480
11	Pants & Capris	143
12	Socks & Hosiery	184
13	Jeans	532
14	Blazers & Jackets	129
15	Shorts	469
16	Tops & Tees	518
17	Plus	172
18	Fashion Hoodies & Sw	488
19	Underwear	306
20	Dresses	213
21	Sleep & Lounge	423
22	Active	358
23	Suits & Sport Coats	235
24	Suits	38
25	Outerwear & Coats	360
26	Pants	312

4. Customer berdasarkan range umur

```
select
        case
                when age between 0 and 18 then 'Remaja'
                when age between 19 and 55 then 'Dewasa'
                else 'Lanjut usia'
        end as range_umur,
        count(age) as jumlah
from users
group by(range_umur)
```

Hasil :

	range_umur	jumlah
1	Dewasa	24,265
2	Remaja	4,606
3	Lanjut usia	9,827

5. daftar bulan 2024 dengan total sale price > 1000

```
select
        extract('month' from t2.created_at) as bulan,
        concat(t1.first_name, ' ', t1.last_name ) as pelanggan,
        sum(sale_price) as sale_price
from users t1
left join order_details t2 on t1.id = t2.user_id
```

```
where extract(year from t2.created_at ) = 2024
group by bulan,pelanggan
having sum(sale_price) > 1000
order by bulan asc
```

Hasil :

	123 bulan	A-Z pelanggan	123 sale_price
1	1	Belinda Shah	1,012.97000027
2	4	Kim Anderson	1,155.77000046
3	6	Anita Nicholson	1,054.8900013
4	6	Jessica Thomas	1,039.99000168
5	7	Shawn Lee	1,015.890007
6	7	Timothy Russell	1,015.89999771
7	8	Janet Cole	1,059.21999931
8	8	Shane Olson	1,059.5
9	8	Shannon Singh	1,022.89999962
10	9	Brittney Tyler	1,014.20999908
11	9	Frank Hunter	1,019
12	9	Joseph Mejia	1,166.93000217
13	9	Michael Hernandez	1,087.99000168
14	10	Carrie Schroeder	1,002.5
15	10	Jeffrey Fisher	1,158.59999847
16	10	Jeffrey Watkins	1,363.9900055
17	10	Joseph Warren	1,013.95000076
18	10	Kristen Horton	1,025.95000076
19	10	Steven Johnson	1,168.23999786
20	11	Christy Moore	1,018.21999931
21	12	David Gonzalez	1,096.29999919
22	12	Gregory Huffman	1,111.26999759
23	12	Julia Mcdaniel	1,041.54000331
24	12	Taylor Smith	1,032.98000141

6. Daftar 10 hari dimana jumlah pemesanan dengan status ’Complete’ terbanyak

```
select
  created_at::date as tanggal,
  count(*) as total_complete
from order_details
where status = 'Complete'
group by created_at::date
order by total_complete desc
limit 10;
```

Hasil :

	tanggal	123 total_complete
1	2024-12-30	89
2	2024-12-26	87
3	2024-12-07	85
4	2024-11-27	82
5	2024-10-15	82
6	2024-11-17	81
7	2024-12-03	81
8	2024-08-23	80
9	2024-12-27	80
10	2024-12-10	79

Reflection

1. Selama menyelesaikan query yang melibatkan filtering, aggregate, dan join, saya memahami bahwa SQL bukan hanya alat untuk mengambil data, tetapi juga berfungsi untuk menjawab kebutuhan bisnis secara langsung. Misalnya, penggunaan group by untuk mengetahui kategori produk dengan penjualan tertinggi dapat

membantu perusahaan menentukan strategi restock, promosi, hingga evaluasi performa produk. Dari proses tersebut, saya melihat bagaimana sebuah query dapat menghasilkan insight yang berdampak nyata pada pengambilan keputusan dalam dunia e-commerce atau retail.

2. Ketika melakukan JOIN antar tabel, saya mempelajari bahwa memahami hubungan antar data dalam database relasional merupakan hal yang sangat penting. Jika struktur tabel dan relasinya tidak dipahami dengan benar, analisis dapat menghasilkan informasi yang salah atau menyesatkan. Misalnya, join yang tidak tepat dapat menyebabkan data terduplikasi atau bahkan hilang, sehingga menghasilkan interpretasi yang keliru mengenai jumlah transaksi atau perilaku pelanggan. Risiko tersebut menunjukkan bahwa pemahaman relasi data sangat diperlukan agar insight yang dihasilkan akurat dan dapat dipercaya.